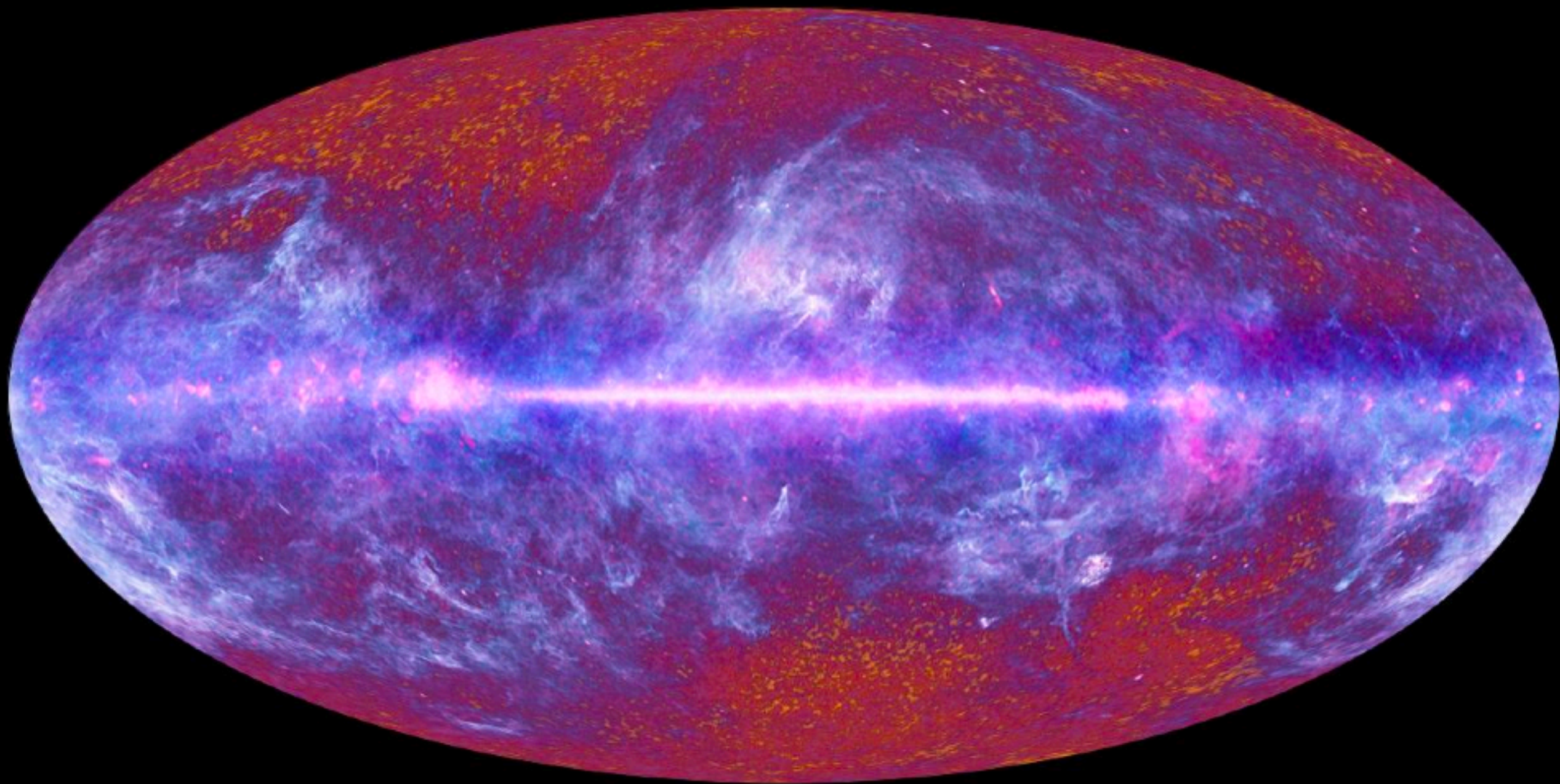
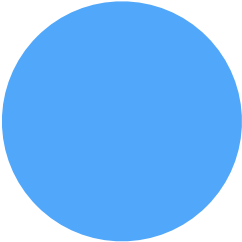
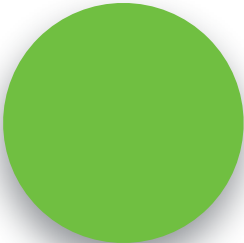




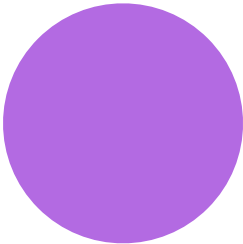


A solid red circle serves as the background for the text.

Bayes'
theorem











Enough about probability (for now). Let's turn now to some results from

contemporary physics which will be important to the argument which

follows.

A first question: what does the theory provide by physics include?

“The standard model of physics presents a theory of the electromagnetic, weak, and strong forces, and a classification of all known elementary particles. The standard model specifies numerous physical laws, but that's not all it does. According to the standard model there are roughly two dozen dimensionless constants that characterize fundamental physical quantities.” (Hawthorne & Isaacs, “Fine-tuning fine-tuning”)



These 'dimensions' will be our focus.

One such constant is the cosmological constant, which measures the

energy density of empty space.

The cosmological constant is just a number which (as far as we know)

could have different values consistent with the laws of physics.

One thing that the standard model of physics gives us is a measure of how

(like the cosmological constant) would fall in a certain range.

likely it is, given the laws of the nature, that the fundamental constants

hard to see how life could have evolved. If there were no planets, it is hard

to see how life could have evolved.

life to have evolved. For example, if there were nothing but hydrogen, it is

We can make certain plausible assumptions about what it would take for

Given assumptions such as these, we can look at what the standard model

constant has a value which would permit the evolution of life.

of physics tells us about how likely it is that, for example, the cosmological

“Physicists have determined the (approximate) values of the fundamental constants by measurement.

(There's no way to derive the values of the fundamental constants from other aspects of the standard model. Any quantities that could be so derived wouldn't be fundamental.) Still, the underlying theory favored some sorts of parameter-values over others. ... Physicists made the startling discovery that -- given antecedently plausibly assumptions about the nature of the physical world -- the probability that a universe with general laws like ours would be habitable was staggeringly low.”

supporting value, given the laws of nature, is very low.

This is the claim that the cosmological constant having a life-

We should emphasize just how small we take the life-permitting parameter values to be according to the physically-respectable measures. “Small” here doesn’t mean “1 in 10,000” or “1 in 1,000,000”. It means the kind of fraction that one would resort to exponents to describe, as in “1 in 10 to the 120”. The kind of package that we have in mind tells us that only a fantastically small range is life permitting.

nature, is



omplimentary constant falling in a life-supporting range, given the laws of

The claim is that, according to current physics, the probability of the

1

10120

This is the claim that the cosmologist has a life-

supporting value, given the law of nature, is very low.

We should emphasize just how small we take the life-permitting parameter values to be according to the physically-respectable measures. “Small” here doesn’t mean “1 in 10,000” or “1 in 1,000,000”. It means the kind of fraction that one would resort to exponents to describe, as in “1 in 10 to the 120”. The kind of package that we have in mind tells us that only a fantastically small range is life permitting.

Let's now ask how to turn these facts about current physics and

probability theory into an argument for the existence of God.

in the turn, so we can consider two different hypotheses about the origins

of the universe.

Much as we considered two different hypotheses about the number of balls

created by an intelligent designer. Let the non-design hypothesis be the

Let the design hypothesis be the hypothesis that the universe

hypothesis that it was not.

value in the life-permitting range.

Let e be the fact that the cosmological constant has a

Then what do we know about relevant probabilities?



$$Pr(e \mid \text{non-design}) = \frac{1}{10^{120}}$$



$$Pr(e | \text{design}) \equiv 0.5$$

Bayes' theorem

$$P(h|e) = \frac{P(h)*P(e|h)}{P(e)}$$

numbers.

design hypothesis given evidence is that just plugging in the

With this information in hand, figuring out the probability of the non-



$$Pr(\text{non-design}) = 0.5$$



$$Pr(\text{design}) = 0.5$$



$$Pr(e) = 0.25$$



$$Pr(\text{non-design} \mid e) = \frac{0.5 * \frac{1}{10^{120}}}{0.25} = \frac{2}{10^{120}}$$

to give you some idea: the odds of winning Powerball are about 1 in 300 million. Now

It is difficult to think about numbers as large as the denominator of this fraction. But



consider the odds of winning Powerball one trillion times in a row. Call that a “super

PowerBar!

that a 'super duper Powerball.'

Now consider the odds of winning a super Powerball one trillion times in a row. Call

The odds of this happening are about $1/10^{44}$ — so much, much higher than the

Now consider the odds of winning a super Powerball one trillion times in a row.

odd of the universe being life-permitting by chance.

superduperpowerball a trillion times in a row.

the non-design hypothesis being true are vastly lower than the chances of winning a

probability of 0.5 to the non-design hypothesis, you should think that the chances of

This means that if you simply take the physics at face value, and begin by assigning a

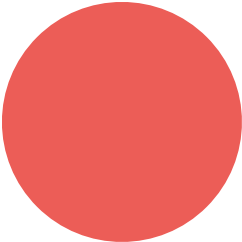
minus the very small number we're just discussing:

By contrast, the probability of the design hypothesis is very close to 1. It is, in fact, 1.



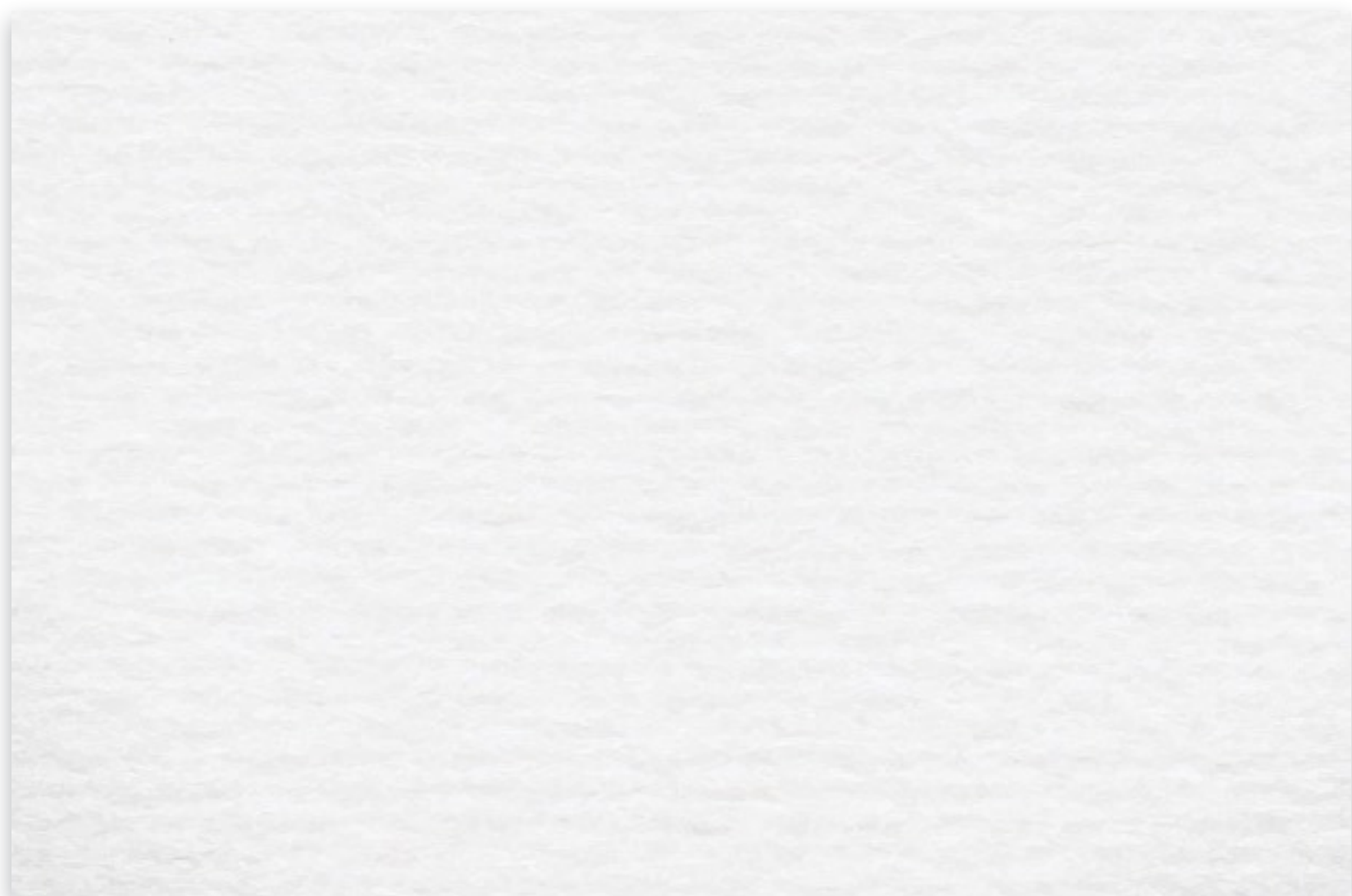
$$Pr(\text{design} \mid e) = 1 - \frac{2}{10^{120}}$$

This about as close to certainty as it is possible to get.





objections



4. Bayes', theorem.

$$3. P(\text{design}) = P(\text{non-design}) = 0.5.$$



5



P

2



8. If the universe was created by an



6. *Life exists.*

$$7. P(\text{design}) \approx 1. \quad (5, 6)$$

$$\text{C.C.}P(\text{God exists}) \approx 1. (7,8)$$

1, 2, 3, 4

independent designer, then God exists.

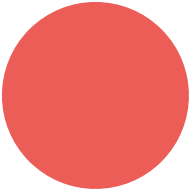
$$(life/death) \equiv 0.5$$

$$(1\text{ife} \mid \text{non-design}) = \frac{1}{10^{120}}$$

$$(\text{design} / \text{life}) \approx 1$$









objections

The claim is that, according to current physics, the probability of the



nature, is

cosmological constant falling in a life-supporting range, giving the laws of

1



10120



the argument



the argument



fine-
tuning of the
universe



fine-
tuning of the
universe